

FSOC Coarse Pointing, Acquisition & Tracking - Performance Report

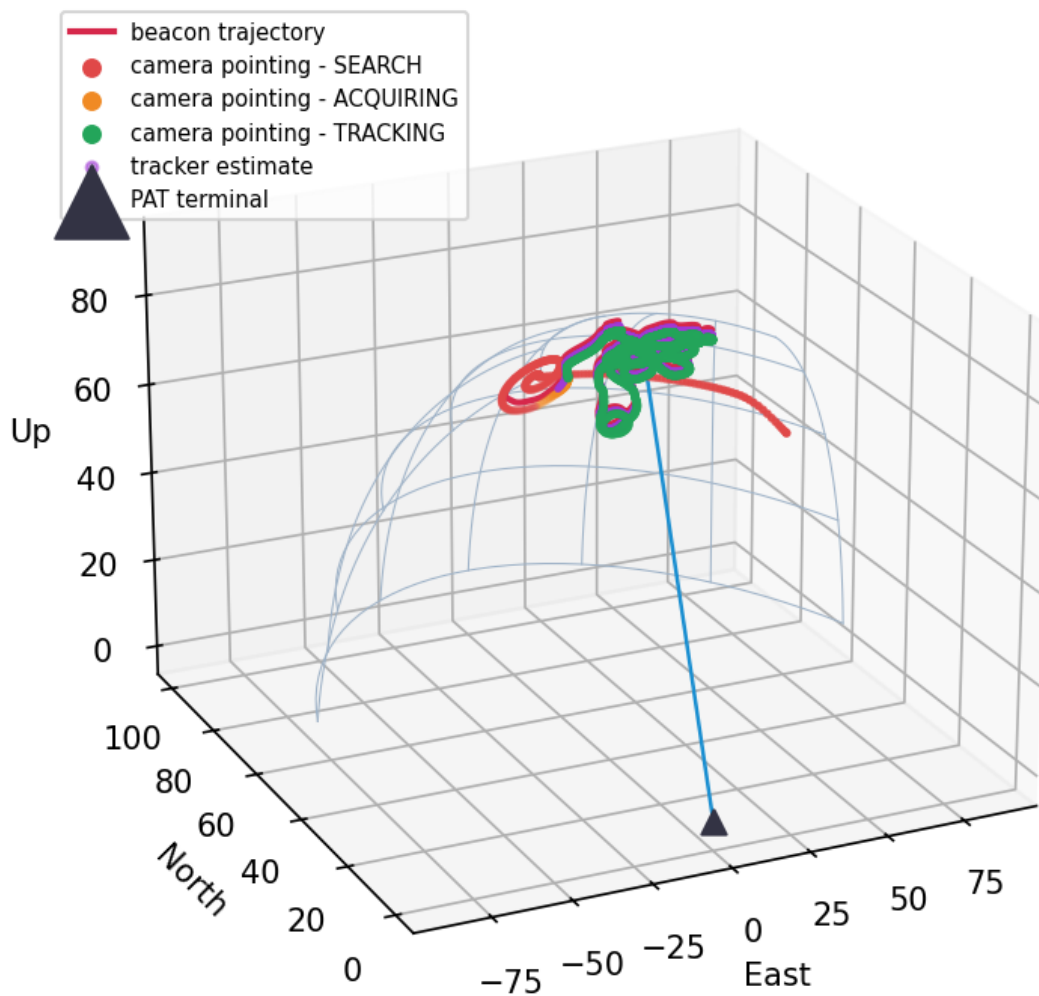
Run	
Generated	2026-10-02T14:51:25
Seed	1400446077
Identification method	hybrid
Beacon path	random_walk at 2.54 deg/s
Decoys	2 (1 blinking)
Turbulence / vibration	1.95 / 1.45
Clouds	no
Reported-position error (GPS prior)	4.15 deg
Camera start	az 53.5, el 0.1 deg (52.2 deg from the reported position)

Performance summary

Metric	Value
Simulation duration	120.0 s
Frames simulated	3600
Achieved frame rate	17.9 FPS
Processing time (mean / max)	5.62 / 22.61 ms
Processing throughput	178 FPS
Acquisition time	8.30 s
Time locked on beacon (whole run)	92.8 %
Lock retention (after first lock)	99.7 %
Lock retention while beacon visible	99.7 %
Lock losses	9
Tracking error mean / RMS / max	5.35 / 6.25 / 17.92 px
Tracking error mean / max	1.166 / 3.910 mrad
Ready for fine-pointing handover	90.5 % of locked time
False-lock frames	11
Candidates rejected / locks dropped	0 / 0
Beacon in field of view / occluded	94.4 / 0.0 %

3D view of the run

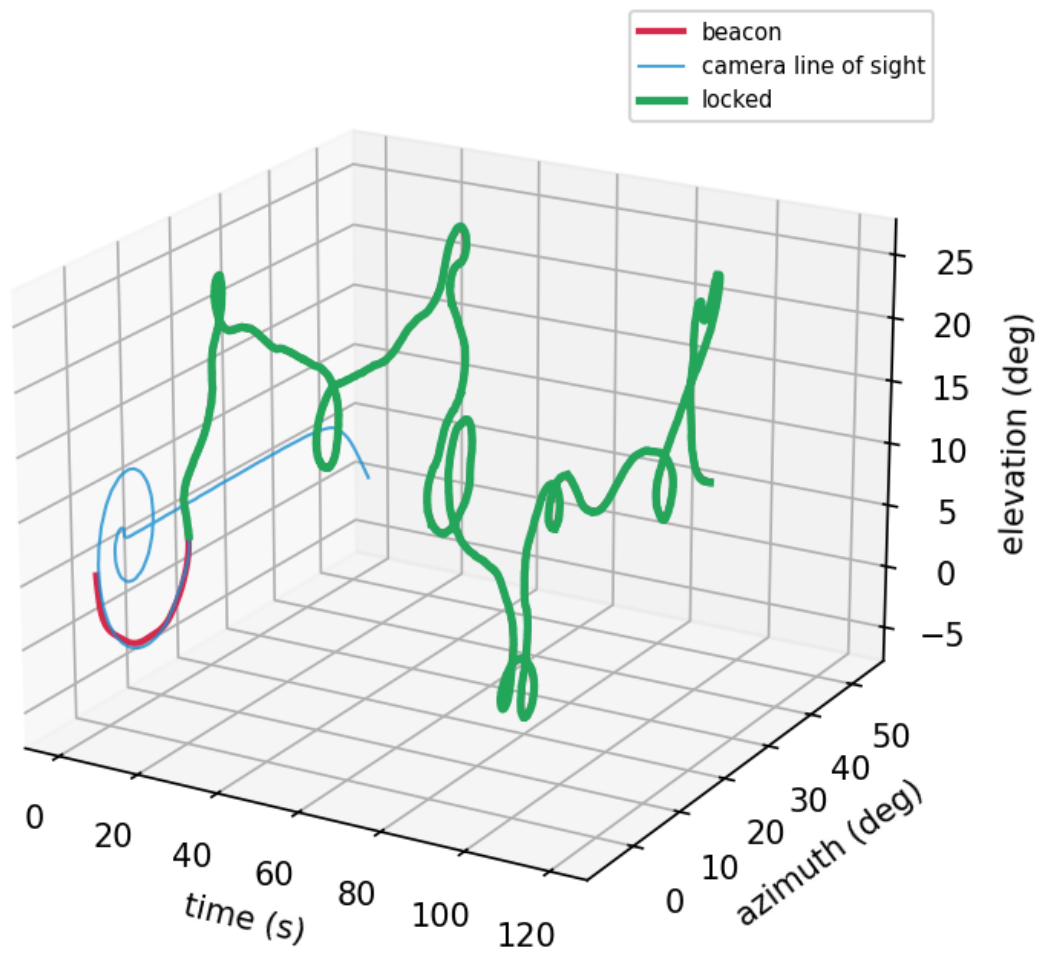
3D view: beacon trajectory and camera line of sight on the sky dome



Red: true beacon path. Dots: where the camera pointed, coloured by tracker state (red search, orange acquiring, green tracking, yellow coasting). Purple: tracker estimate.

3D view: azimuth / elevation over time

3D view: azimuth / elevation over time - beacon vs camera pointing



The camera line of sight (blue, green while locked) converging onto and following the beacon (red) shows acquisition and tracking in one picture.

Time charts

Tracking performance over time

